

S. No	Title (Name of the journal, author and publication details)	Methodology (Provide a Summary of key studies and their findings)	Identification of gaps and limitations (Identify the limitations of the Research Paper)
1	A machine learning-based universal outbreak risk prediction tool; Computers in Biology and Medicine; Zhang, T., Rabhi, F., Chen, X., Paik, H., & MacIntyre, C. R. (2024)	- Utilized outbreak data from 43 diseases across 206 countries to build a universal system. - Employed an ensemble of five machine learning models: Neural Network, XGBoost, LogitBoost, Random Forest, and Kernel SVM. - Achieved 80%–90% prediction accuracy by analyzing economic, cultural, social, and epidemiological factors.	- Performance varies across different datasets and realistic scenarios. - While universal, specific local epidemiological nuances might still challenge model precision.
2	Predicting infectious disease for biopreparedness and response: A systematic review of machine learning and deep learning approaches; One Health; Keshavamurthy, R., Dixon, S., Pazdernik, K. T., & Charles, L. E. (2022)	- Conducted a PRISMA systematic review of 237 articles published between 2001 and 2021. - Found that tree-based ML (38.4%) and feed-forward neural networks (26.6%) were the most frequent approaches. - Identified that most studies focused on temporal incidence (66.7%) and disease risk (38.0%).	- Significant geographic bias, with 86.5% of articles focusing on human diseases and limited representation of zoonotic hosts (5.1%). - Less than 1% of reviewed studies addressed uncertainty quantification or computational efficiency.
3	COVID-19 Outbreak Prediction with Machine Learning; medRxiv (Preprint); Ardabili, S. F., Mosavi, A., Ghamisi, P., et al. (2020)	- Performed a comparative analysis of ML and soft computing models for COVID-19. - Investigated Multi-Layered Perceptron (MLP) and Adaptive Network-based Fuzzy Inference System (ANFIS). - Proposed ML as a more effective tool than standard epidemiological models for handling the nonlinear, complex nature of the outbreak.	- Standard models showed low accuracy for long-term prediction due to high uncertainty and lack of essential data. - Generalization and robustness of existing models need significant improvement for diverse geopolitical areas.
4	Disease Prediction Using Machine Learning Over Big Data; Computer Science & Engineering: An International Journal (CSEIJ); Vinitha S, Sweetlin S, Vinusha H, & Sajini S (2018)	- Proposed a system using Machine Learning Decision Tree and Map Reduce algorithms. - Used a latent factor model to rebuild missing or incomplete medical data to improve study exactness. - Reached a calculation accuracy of 94.8% on real-life hospital data for cerebral infarction.	- Incomplete data quality in biomedical communities significantly reduces the exactness of predictions. - Unique regional disease appearances can weaken the generalizability of outbreak predictions.
5	Google Search Trends Predicting Disease Outbreaks: An Analysis from India; Healthcare Informatics Research; Verma, M., Kishore, K., Kumar, M., et al. (2018)	- Assessed the temporal correlation between Google Trends and Integrated Disease Surveillance Programme (IDSP) data. - Extracted top terms for four febrile illnesses (e.g., Dengue) using Google Correlate. - Aimed to determine the feasibility of using internet search data as a "nowcasting" tool for prompt detection.	- Google Trends data provides "real-time" insights but does not replace the precision of clinical surveillance. - Data can be influenced by public concern and media coverage rather than actual disease incidence.

6	Correlation between Google Trends on dengue fever and national surveillance report in Indonesia; Global Health Action; Husnayain, A., Fuad, A., & Lazuardi, L. (2019)	- Conducted a quantitative study using time series data (2012–2016) comparing Google Trends with official dengue reports. - Used Moving Average analysis, Pearson correlation, and Time lag correlations to measure relationships. - Found high correlation (R-value 0.921 to 0.937) between search terms and actual cases.	- Digital traces are sensitive to public reaction and media interest, which may not always reflect clinical reality. - Limited by the specific time frame (2012–2016) and regional internet penetration.
7	Real-Time Epidemic Forecasting: Challenges and Opportunities; Health Security; Desai, A. N., Kraemer, M. U. G., et al. (2019)	- Discussed data needs in surveillance, mobility, environmental susceptibility, and healthcare capacity. - Evaluated the use of stochastic semi-mechanistic models and machine learning for refining future forecasts. - Highlighted the importance of real-time digital disease surveillance and open data sharing.	- Standardized case definitions and timely data sharing constraints limit the precision of predictive models. - Resource-limited settings lack the granular data necessary for accurate forecasting.
8	Computational Forecasting Methodology for Acute Respiratory Infectious Disease Dynamics; Int. J. Environ. Res. Public Health; González-Bandala, D. A., et al. (2020)	- Proposed a methodology merging machine learning, a projection model, and a smoothed endemic channel calculation. - Analyzed weekly acute respiratory infection (ARI) data in Mexico alongside Google search terms. - Achieved a Maximum Absolute Percent Error (MAPE) of 21.7%.	- Previous research lacks focus on the specifics of data retrieval, preprocessing, and feature extraction. - Models are often specific to certain time periods and may lack flexibility for different seasonal behaviors.
9	Data analytics for predicting disease outbreaks: A review of models and tools; Int. J. Life Science Research Updates; Adegoke, B. O., et al. (2024)	- Reviewed statistical approaches, machine learning models, and epidemiological frameworks. - Investigated tools like GIS, data visualization, and big data analytics platforms. - Analyzed the role of social media and web data in improving prediction timeliness.	- Encountered hurdles in model complexity, interpretability, and ethical considerations. - Data quality and availability remain major barriers to successful real-world application.
10	Predictive Modeling for Disease Outbreaks: A Review of Data Sources and Accuracy; International Medical Science Research Journal; Ijeh, S., et al. (2024)	- Explored predictive modeling focusing on epidemiological, environmental, and mobility data. - Evaluated various modeling approaches from statistical to network analysis and AI/IoT integration. - Emphasized the critical role of validation and accuracy metrics in public health decision-making.	- High model complexity can lead to challenges in practical implementation by health officials. - Ethical concerns regarding the use of personal mobility and social media data for surveillance.
11	Predicting Infectious Disease Outbreaks with Machine Learning and Epidemiological Data; Journal of Advanced Zoology; Rahman, S. Z., et al. (2023)	- Examined the shift in public health toward using machine learning (ML) and epidemiological data for forecasting. - Analyzed the integration of diverse datasets, including environmental and clinical data, into predictive frameworks. - Found that ML methodologies significantly improve the ability to control and mitigate infectious outbreaks compared to traditional methods.	- High computational costs and the need for specialized expertise can limit accessibility for some health departments. - Ensuring data privacy and security remains a significant challenge when handling sensitive epidemiological information.
12	Identification and Prediction of Chronic Diseases Using Machine Learning Approach; Journal of Healthcare Engineering; Alanazi, R. (2022)	- Focused on identifying common chronic illnesses at early stages using cutting-edge ML techniques. - Utilized both structured and unstructured medical data to ensure reliable patient categorization. - Proposed that early recognition via automated systems is crucial for preventing disease extremity and improving patient care.	- Accuracy is highly dependent on the quality of the input medical data, which can often be incomplete. - Manual identification by doctors is still sometimes necessary to confirm complex or borderline cases.

13	The roles of machine learning methods in limiting the spread of deadly diseases: A systematic review; Heliyon; Alfred, R., & Obit, J. H. (2021)	- Conducted a systematic review of ML algorithms (e.g., ARMA, MLP, SVM) and datasets used for disease prediction and detection. - Identified that unstructured data, such as social media and search keywords, are frequently used for forecasting. - Outlined various ensemble and individual model performances across diverse deadly disease outbreaks.	- Many existing reviews fail to discuss the specific datasets and performance measurements used in detail. - Challenges remain in scaling these models for real-time global deployment across different infectious diseases.
14	Predicting COVID-19 Incidence Through Analysis of Google Trends Data in Iran; JMIR Public Health and Surveillance; Ayyoubzadeh, S. M., et al. (2020)	- Conducted a pilot study using data mining and deep learning on Google Trends data in Iran. - Analyzed search engine queries to provide insights into population behavior during the COVID-19 epidemic. - Investigated the feasibility of using internet-based resources to manage and predict incidence rates.	- Google Trends data may be influenced by media-driven panic rather than actual clinical symptoms. - Limited to populations with high internet access, potentially biasing results against certain demographics.
15	On the predictability of infectious disease outbreaks; Nature Communications; Scarpino, S. V., & Petri, G. (2019)	- Adopted permutation entropy as a model-independent measure to study the fundamental limits of disease predictability. - Analyzed a diverse collection of outbreak time series to identify "entropy barriers" for forecasting. - Found that social network heterogeneity and shifting model structures significantly influence forecast horizons.	- Identified an entropy barrier that can limit long-term time series forecasting for complex adaptive systems. - Model selection across long time series remains challenging due to the dynamic nature of outbreaks.
16	Technology to advance infectious disease forecasting for outbreak management; Nature Communications; George, D. B., et al. (2019)	- Discussed the integration of forecasting into decision-making for public health response. - Evaluated how modern technologies can accelerate adoption of forecasting to save lives and reduce economic impact. - Highlighted the role of medical countermeasures (vaccines, diagnostics) versus traditional public health measures.	- Persistent "data gaps" undermine the ability to target resources and track accountability effectively. - Public health practitioners may lack the tools or training to fully integrate complex analytics into rapid response.
17	Prediction of dengue outbreak in Selangor Malaysia using machine learning techniques; Scientific Reports; Salim, N. A. M., et al. (2021)	- Evaluated five districts in Selangor using ML models (e.g., SVM with linear kernel) to predict dengue outbreaks. - Utilized climate variables (temperature, humidity, rainfall) and the "week-of-the-year" as key predictors. - Achieved 70% accuracy with the SVM model after addressing data imbalances.	Initial models struggled with imbalanced data, requiring specific adjustments to improve sensitivity. The study is localized to one state, meaning results may not be directly applicable to other regions with different climates.